Universal Stability Theorem for On-Shell Flow-Adaptive Networks

The Green map is the shifted inverse $M_w = (L_w + J/|V|)^{-1}$, which on $\mathbf{1}^\perp$ agrees with L_w^+. There is no matrix Moore–Penrose $^+$ in Mathlib 4.28 (Penrose 1955).

Kernel (`lake build UniversalStability`, Lean / Mathlib 4.28.0, **no `sorry`**): Definitions through Theorem 5, including Hessian spectral decomposition, modal Jury stability, Stein series $\mathcal{P} \succeq I$, C^2 Taylor bound on F_c, quadratic margin estimation, and the full discrete nonlinear leapfrog contraction with a forward-invariant ellipsoidal basin of attraction (`theorem5_local_invariance`).

Artifact & Formal Verification. All definitions, lemmas, and theorems through Theorem 5 (including the modal Jury decomposition, Stein Lyapunov solver, and the discrete nonlinear ellipsoid invariance theorem `theorem5_local_invariance`) have been machine-checked in Lean 4 / Mathlib (toolchain `v4.28.0`). The default build target contains **zero `sorry`** and depends strictly on the three standard foundational axioms (`propext`, `Classical.choice`, `Quot.sound`). Non-vacuity of the theorem hypotheses is demonstrated via an explicit 2-node path instance (`twoNode_theorem5`): at $c = 0$, force balance is $V'(w^\star) = 0$. (Note $V'(1) = -6 \neq 0$; the unique positive root of V' satisfies $w - 1 = w^{-3}$ and lies in $(1, \infty)$, hence is admissible.) The full artifact, CI pipeline (`./scripts/verify.sh`), and reproduction scripts are at https://github.com/tripstoph/universal-stability-theorem.

Provenance. This repository—including the manuscript, the Lean 4 formalization, CI, and all supporting files—was generated by AI (Cursor language-model agents Cursor Grok 4.6 High & Gemini 3.7 Flash) under human direction of the mathematical program.

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1. Definitions

Let V, E be finite nonempty types. Let $B \in \mathbb{R}^{V \times E}$ satisfy

$$B^T \mathbf{1} = 0 \qquad \text{(balanced incidence)}$$

and $\ker B^T \subseteq \operatorname{span}\{\mathbf{1}\}$ (connectedness). Let $w \in \mathbb{R}^E_{>0}$ and $W = \operatorname{diag}(w)$. The weighted Laplacian is $L_w = BWB^T$ (Kirchhoff 1847; Chung 1997). Then $L_w \Psi = 0$ if and only if $B^T \Psi = 0$, hence $\ker L_w = \operatorname{span}\{\mathbf{1}\}$.

Let $\eta \in \mathbb{R}^V$ with $\sum_v \eta_v = 0$. Write J for the all-ones matrix on V and

$$M_w := L_w + \frac{1}{|V|} J.$$

For $w > 0$ and connected B, M_w is positive definite, hence invertible. The on-shell potential is

$$\Psi(w) := M_w^{-1} \eta.$$

It is the unique mean-zero solution of $L_w \Psi = \eta$. Edge drops are $\Delta\Psi(w) = B^T \Psi(w)$.

Constitutive potential on $(0, \infty)$:

$$V(w) = 3(w-1)^2 + 3w^{-2}, \qquad V'(w) = 6(w-1) - 6w^{-3}, \qquad V''(w) = 6 + 18w^{-4}.$$

One has $V'' > 0$, so V' is a bijection $(0, \infty) \to \mathbb{R}$. Set

$$h(w) := V''(w) + \frac{2V'(w)}{w} = 18 - \frac{12}{w} + \frac{6}{w^4}.$$

For all $w > 0$,

$$h(w) \geq 18 - 9 \cdot 2^{-1/3} > 10.$$

The bound is attained at $w = 2^{1/3}$.

For $c \geq 0$,

$$F_c(w)_e = \frac{c}{2}(\Delta\Psi_e(w))^2 + V'(w_e), \qquad \Phi(w) = \sum_{e \in E} V(w_e) - \frac{c}{2} \eta^T M_w^{-1} \eta.$$

An on-shell equilibrium is a point $w^\star \in \mathbb{R}^E_{>0}$ with $F_c(w^\star) = 0$. Write $\Omega := \mathbb{R}^E_{>0}$.

The algebraic Hessian at drops $d = \Delta\Psi(w)$ is the matrix

$$\mathcal{H}_w = -c\, D_d\, (B^T M_w^{-1} B)\, D_d + \operatorname{diag}(V''(w)),$$

where $D_d = \operatorname{diag}(d)$.

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2. Theorem 1 (Fréchet derivative of F_c)

On Ω, F_c is Fréchet differentiable (Dieudonné 1960) and

$$DF_c(w)[\delta w] = \mathcal{H}_w\, \delta w.$$

Proof

Differentiating $M_w \Psi(w) = \eta$ (equivalently $L_w \Psi = \eta$ on mean-zero vectors) in the direction δw yields

$$D\Psi(w)[\delta w] = -M_w^{-1} B(d \odot \delta w),$$

where $\odot$ is the Hadamard product. Differentiating $F_c(w)_e = \frac{c}{2} d_e^2 + V'(w_e)$ then produces the displayed matrix. ■

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3. Lemma 1 (transfer Rayleigh bound)

For all $x \in \mathbb{R}^E$,

$$x^T (B^T M_w^{-1} B) x \le x^T W^{-1} x.$$

Equivalently, writing $\widetilde{B} = BW^{1/2}$ and $\Pi = \widetilde{B}^T M_w^{-1} \widetilde{B}$, one has that Π is an orthogonal projector, $B^T M_w^{-1} B = W^{-1/2} \Pi W^{-1/2}$, and $0 \preceq \Pi \preceq I$.

Proof

Completing the square (Thomson's principle (Doyle and Snell 1984)) gives the Rayleigh inequality for any right inverse of L_w on $\operatorname{im} B$. The identification with Π is the change of variables $y = W^{-1/2} x$. Orthogonal projectors satisfy $0 \le y^T \Pi y \le \|y\|^2$. ■

Corollary 1

For all $x \in \mathbb{R}^E$,

$$x^T \mathcal{H}_w x \ge \sum_e \Big(V''(w_e) - \frac{c\, d_e^2}{w_e} \Big) x_e^2.$$

(The interaction term is negative semidefinite because $c \ge 0$ and $B^T M_w^{-1} B$ is positive semidefinite.)

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4. Theorem 2 (equilibrium Hessian floor)

Let $w^\star \in \Omega$ satisfy $F_c(w^\star) = 0$. Then for all $x \in \mathbb{R}^E$,

$$x^T \mathcal{H}_{w^\star} x \ge (18 - 9 \cdot 2^{-1/3}) \|x\|^2.$$

Proof

Force balance on edge e is $c\, d_e^2 = -2V'(w_e^\star)$. Substituting into Corollary 1 produces

$$x^T \mathcal{H}_{w^\star} x \geq \sum_e h(w_e^\star)\, x_e^2.$$

The scalar bound $h \geq 18 - 9 \cdot 2^{-1/3}$ finishes the estimate. ■

This is a quadratic-form lower bound. It is not, in the kernel, a second-derivative test: IsLocalMin Φ at $w^\star$ is not discharged.

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5. Lemma 2 (gradient structure)

On Ω,

$$D\Phi(w)[\delta w] = \langle F_c(w), \delta w \rangle.$$

Thus $F_c = \nabla \Phi$ and $DF_c = D^2\Phi$ as Fréchet derivatives.

Proof

The map $w \mapsto \eta^T M_w^{-1} \eta$ has directional derivative $-\sum_e \delta w_e\, d_e^2$. Differentiating $\sum V(w_e)$ contributes $V'(w_e)$. ■

A local minimizer of Φ on Ω is therefore an on-shell equilibrium.

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6. Theorem 2A (coercivity and existence)

Let $y := B^T M_1^{-1} \eta$ be the unit-weight Kirchhoff current (Kirchhoff 1847), and write $C := \sum_e y_e^2 \geq 0$, $w_{\text{min}} := \min_e w_e$, $w_{\text{max}} := \max_e w_e$. Then

$$\Phi(w) \geq \sum_e V(w_e) - \frac{c}{2} \sum_e \frac{y_e^2}{w_e} \geq 3w_{\text{min}}^{-2} - \frac{cC}{2\, w_{\text{min}}},$$

and likewise $\Phi(w) \geq 3(w_{\text{max}} - 1)^2 - \frac{cC}{2\, w_{\text{min}}}$. Hence $\Phi(w) \to +\infty$ as $w_{\text{min}} \to 0^+$ or $w_{\text{max}} \to +\infty$.

Every sublevel $\{w \in \Omega : \Phi(w) \leq K\}$ is compact in Ω. Therefore Φ attains a global minimizer $w^\star \in \Omega$, and $F_c(w^\star) = 0$.

Proof

Thomson's principle (Doyle and Snell 1984) supplies $\eta^T M_w^{-1} \eta \leq \sum_e y_e^2 / w_e \leq C / w_{\text{min}}$. The two lower bounds on V are $V(w) \geq 3w^{-2}$ and $V(w) \geq 3(w-1)^2$. Thresholds $\varepsilon \in (0, 1]$ and $R \geq 1$ may be chosen so that $\Phi > K$ off the box $[\varepsilon, R]^E$. That box is compact and lies in Ω; Φ is continuous there, so a minimizer exists. It is a local minimizer in the open set Ω, hence a zero of F_c. ■

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7. Theorem 2B (uniqueness)

There is a unique on-shell equilibrium. Equivalently, Φ has a unique critical point on Ω, and it is the global minimizer of Theorem 2A.

Proof

For each drop $t \in \mathbb{R}$ there is a unique $w(t) > 0$ with $V'(w(t)) = -(c/2)t^2$. The current $\sigma(t) := w(t)\, t$ satisfies

$$\sigma'(t) = w(t) - \frac{ct^2}{V''(w(t))} = \frac{w(t)\, h(w(t))}{V''(w(t))} > 0,$$

so σ is strictly increasing.

At equilibrium, $w_e = w(\Delta\Psi_e)$ and $I_e = \sigma(\Delta\Psi_e)$ obeys $BI = \eta$. If w, w' are two equilibria with drops δ, δ', then $B(I - I') = 0$, hence

$$\sum_e (\sigma(\delta_e) - \sigma(\delta_e'))(\delta_e - \delta_e') = \langle I - I', B^T(\Psi - \Psi') \rangle = \langle B(I - I'), \Psi - \Psi' \rangle = 0.$$

Each summand is nonnegative, so each is zero, so $\delta = \delta'$, so $w = w'$.

A critical point of Φ is a zero of F_c by Lemma 2. Thus there is at most one critical point. Combined with Theorem 2A it is the global minimizer. ■

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8. Discrete linearization

The damped leapfrog map (Verlet 1967; Hairer et al. 2006) is

$$w_{t+1} = w_t + \Delta T\, v_t, \qquad v_{t+1} = (1 - \gamma\Delta T)v_t - \Delta T\, F_c(w_{t+1}).$$

Its linearization at $(w^\star, 0)$ is the block matrix on $\mathbb{R}^E \oplus \mathbb{R}^E$

$$\mathcal{J} = \begin{pmatrix} I & \Delta T\, I \\ -\Delta T\, \mathcal{H}_{w^\star} & (1 - \gamma\Delta T)I - (\Delta T)^2 \mathcal{H}_{w^\star} \end{pmatrix}.$$

This is the linearization of $\mathcal{T}$ at $(w^\star, 0)$. Global invariance of Ω under $\mathcal{T}$ is not claimed; local invariance is Theorem 5.

On $\{w_e \geq 1/3\}$ one has, for all x,

$$(18 - 9 \cdot 2^{-1/3})\|x\|^2 \leq x^T \mathcal{H}_w x \leq 1464\, \|x\|^2.$$

The upper bound uses Corollary 1's sign on the interaction together with $V''(w) \leq V''(1/3) = 1464$. The constraint $w_e \geq 1/3$ is an additional hypothesis on the equilibrium (smallness of η); it is not implied by $F_c(w^\star) = 0$.

If $\mathcal{H} v = \lambda v$, the restriction of $\mathcal{J}$ to span$\{(v, 0), (0, v)\}$ is

$$J_\lambda = \begin{pmatrix} 1 & \Delta T \\ -\Delta T\, \lambda & 1 - \gamma\Delta T - \lambda(\Delta T)^2 \end{pmatrix}.$$

The characteristic polynomial is $\mu^2 - \tau\mu + \delta = 0$ with $\tau = 2 - \gamma\Delta T - \lambda(\Delta T)^2$ and $\delta = 1 - \gamma\Delta T$. The Jury inequalities $|\delta| < 1, 1 - \tau + \delta > 0, 1 + \tau + \delta > 0$ (Jury 1962) are equivalent to

$$|1 - \gamma\Delta T| < 1, \qquad \lambda(\Delta T)^2 > 0, \qquad 4 - 2\gamma\Delta T - \lambda(\Delta T)^2 > 0.$$

Theorem 3 (modal Jury)

Let $\gamma = 3$ and $0 < \lambda \leq 1464$. If

$$0 < \Delta T < \Delta T_\star := \frac{-3 + \sqrt{5865}}{1464},$$

then the three Jury inequalities hold for J_λ. Consequently $\rho(J_\lambda) < 1$: every real characteristic root satisfies $|\mu| < 1$; if the discriminant is negative, the complex roots satisfy $|\mu|^2 = 1 - 3\Delta T < 1$.

If a matrix $P \in \mathbb{R}^{2\times 2}$ solves $J^T P J - P = -I_2$, then

$$\mathcal{V}(Jx) = \mathcal{V}(x) - \|x\|^2$$

for $\mathcal{V}(x) = x^T P x$, and $\mathcal{V}(Jx) < \mathcal{V}(x)$ for $x \neq 0$ (Lyapunov 1992; Stein 1952). If in addition $\mathcal{V}(y) \leq \lambda_{\max}\|y\|^2$ with $\lambda_{\max} > 0$, then $\mathcal{V}(Jx) \leq (1 - \lambda_{\max}^{-1})\mathcal{V}(x)$.

Proof

$|1 - 3\Delta T| < 1$ holds for $\Delta T < 2/3$, which is weaker than $\Delta T_\star$. The middle Jury inequality is $\lambda(\Delta T)^2 > 0$. The third is most restrictive at $\lambda = 1464$ and rearranges to the positive root of $1464\,t^2 + 6t - 4 = 0$.

A real root of $\mu^2 - \tau\mu + \delta = 0$ with Jury's inequalities cannot satisfy $|\mu| \geq 1$. Underdamped roots are conjugates with product δ, hence $|\mu|^2 = \delta = 1 - 3\Delta T$. A double real root is the discriminant-zero case of the real-root statement. Thus every eigenvalue of J_λ in $\mathbb{C}$ has modulus strictly less than one. ■

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9. Theorem 4 (spectral radius of $\mathcal{J}$; Stein equation)

Let $\gamma = 3$, $0 < \Delta T < \Delta T_\star$, and $w^\star \in \Omega$ with $F_c(w^\star) = 0$ and $w_e^\star \geq 1/3$ for all e. Write $n := |E|$ and $\mathcal{H} := \mathcal{H}_{w^\star}$. Then

$$\rho(\mathcal{J}) < 1,$$

and there exists a unique symmetric $\mathcal{P} \succ 0$ on $\mathbb{R}^{2n}$ solving

$$\mathcal{J}^T \mathcal{P} \mathcal{J} - \mathcal{P} = -I_{2n}.$$

Moreover $\mathcal{P} = \sum_{k\geq 0} (\mathcal{J}^T)^k \mathcal{J}^k \succeq I_{2n}$.

Proof

The matrix $\mathcal{H}$ is real symmetric. Theorem 2 and the bound on $\{w_e \geq 1/3\}$ give, for all $x \in \mathbb{R}^n$,

$$(18 - 9 \cdot 2^{-1/3})\|x\|^2 \leq x^T \mathcal{H} x \leq 1464\|x\|^2.$$

Hence $\operatorname{spec}(\mathcal{H}) \subseteq [18 - 9 \cdot 2^{-1/3}, 1464] \subset (0, 1464]$. Let $Q \in \mathrm{O}(n)$ satisfy $Q^T \mathcal{H} Q = \Lambda = \operatorname{diag}(\lambda_1, \dots, \lambda_n)$ (Horn and Johnson 2013). Set

$$\mathcal{S} := \begin{pmatrix} Q & 0 \\ 0 & Q \end{pmatrix} \in \mathrm{O}(2n).$$

Then

$$\mathcal{S}^T \mathcal{J} \mathcal{S} = \begin{pmatrix} I & \Delta T\, I \\ -\Delta T\, \Lambda & (1 - \gamma\Delta T) I - (\Delta T)^2 \Lambda \end{pmatrix}.$$

Let π be the permutation of coordinates $(x, y) \mapsto (x_1, y_1, \dots, x_n, y_n)$, and $\mathcal{U} := \mathcal{S} P_\pi \in \mathrm{O}(2n)$. Then

$$\mathcal{U}^T \mathcal{J} \mathcal{U} = \bigoplus_{i=1}^{n} J_{\lambda_i}.$$

Similarity preserves the spectrum, and the spectrum of a direct sum is the union of the spectra, so

$$\operatorname{spec}(\mathcal{J}) = \bigcup_{i=1}^{n} \operatorname{spec}(J_{\lambda_i}).$$

Theorem 3 yields $\rho(J_{\lambda_i}) < 1$ for each i, hence $\rho(\mathcal{J}) = \max_i \rho(J_{\lambda_i}) < 1$.

Stein's theorem (Stein 1952): if $\rho(A) < 1$ and $Q \succ 0$, the series $P = \sum_{k\geq 0} (A^T)^k Q A^k$ converges in any matrix norm (Gelfand's formula supplies the geometric bound on $\|A^k\|$; (Rudin 1991)), is the unique symmetric solution of

$A^T P A - P = -Q$, and satisfies $P \succ 0$. For $A = \mathcal{J}$ and $Q = I_{2n}$ one obtains $\mathcal{P}$ as displayed, and the $k = 0$ term gives $\mathcal{P} \succeq I_{2n}$.

The same $\mathcal{P}$ admits the modal construction: for each i, $P_i = \sum_{k \geq 0} (J_{\lambda_i}^T)^k J_{\lambda_i}^k$ solves the 2×2 Stein equation, and

$$\mathcal{P} = \mathcal{U}\Big(\bigoplus_{i=1}^{n} P_i\Big)\mathcal{U}^T.$$

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10. Theorem 5 (local invariance and exponential stability)

Let $\gamma = 3$, $0 < \Delta T < \Delta T_\star$, and let $w^\star$ be as in Theorem 4. Write $z_t = (w_t - w^\star,\, v_t) \in \mathbb{R}^{2n}$ and let $\mathcal{T}$ be the leapfrog map of §8, defined whenever $w_t + \Delta T v_t \in \Omega$. Let $\mathcal{P}$ be the Stein matrix of Theorem 4, $\lambda_M := \|\mathcal{P}\|_2$, $\kappa := \|\mathcal{J}\|_2$, and $\mathcal{V}(z) := z^T \mathcal{P} z$. The Stein floor $\mathcal{P} \succeq I$ gives $\lambda_M \geq 1$ and $\|z\|^2 \leq \mathcal{V}(z)$.[1]

There exists $r_0 > 0$ such that the ellipsoid $\mathcal{E} := \{z : \mathcal{V}(z) \leq r_0^2\}$ satisfies: 1. $\mathcal{E} \subset \Omega \times \mathbb{R}^n$; 2. if $z_0 \in \mathcal{E}$, then $w_0 + \Delta T v_0 \in \Omega$, $z_{t+1} = \mathcal{T}(w_t, v_t) - (w^\star, 0)$ is defined for all $t \in \mathbb{N}$, and $z_t \in \mathcal{E}$; 3. $\|z_t\| \leq \sqrt{\lambda_M}\, q^{t/2}\|z_0\|$, where $q = 1 - \frac{1}{2}\lambda_M^{-1}$.

Proof

Let $w^\star_{\text{min}} := \min_e w^\star_e > 0$ and fix $\rho \in (0, w^\star_{\text{min}})$. The closed Euclidean ball $\overline{B}_\rho(w^\star)$ lies in Ω. On Ω, F_c is C^∞ (inverse of the positive-definite matrix M_w, polynomial in the drops). Hence

$$M_2 := \sup_{w \in \overline{B}_\rho(w^\star)} \|D^2 F_c(w)\|_{\text{op}} < \infty.$$

Taylor with remainder about $w^\star$ (Cartan 1971), using $F_c(w^\star) = 0$ and $DF_c(w^\star) = \mathcal{H}$, gives

$$F_c(w^\star + h) = \mathcal{H}h + \mathcal{R}_F(h), \qquad \|\mathcal{R}_F(h)\|_\infty \leq \tfrac{1}{2}M_2\|h\|_\infty^2$$

whenever $\|h\|_\infty \leq \rho$. Euclidean length satisfies $\|h\|_\infty \leq \|h\|_2 \leq \sqrt{n}\|h\|_\infty$. If $w_{t+1} = w_t + \Delta T v_t$, then $w_{t+1} - w^\star = \delta w_t + \Delta T v_t$ and

$$z_{t+1} = \mathcal{J} z_t + \mathcal{R}(z_t), \qquad \mathcal{R}(z) = \begin{pmatrix} 0 \\ -\Delta T\, \mathcal{R}_F(\delta w + \Delta T v) \end{pmatrix},$$

provided $\|\delta w + \Delta T v\|_2 \leq \rho$. The first block row of $\mathcal{J}$ is $(I, \Delta T I)$, with operator norm $\sqrt{1 + (\Delta T)^2}$, so

$$\|\delta w + \Delta T v\|_2 \leq \sqrt{1 + (\Delta T)^2}\, \|z\|_2.$$

Thus, writing $C_R := \frac{1}{2}\Delta T M_2 (1 + (\Delta T)^2)\sqrt{n}$ (the $\sqrt{n}$ converting the C^2 remainder from the coordinate sup-norm to Euclidean length), one has $\|\mathcal{R}(z)\|_2 \leq C_R\|z\|_2^2$ whenever $\sqrt{1 + (\Delta T)^2}\, \|z\|_2 \leq \rho$.

Expanding the Stein identity $\mathcal{J}^T \mathcal{P} \mathcal{J} - \mathcal{P} = -I$,

$$\mathcal{V}(\mathcal{J}z + \mathcal{R}) = \mathcal{V}(z) - \|z\|_2^2 + 2z^T \mathcal{J}^T \mathcal{P} \mathcal{R} + \mathcal{R}^T \mathcal{P} \mathcal{R}.$$

Hence

$$\left|2z^T \mathcal{J}^T \mathcal{P} \mathcal{R}\right| \leq 2\kappa\lambda_M \|z\|_2\, \|\mathcal{R}\|_2, \qquad \mathcal{R}^T \mathcal{P} \mathcal{R} \leq \lambda_M \|\mathcal{R}\|_2^2,$$

and, on the remainder regime,

$$\mathcal{V}(z_{t+1}) - \mathcal{V}(z_t) \leq -\|z_t\|_2^2\Big(1 - 2\kappa\lambda_M C_R \|z_t\|_2 - \lambda_M C_R^2 \|z_t\|_2^2\Big).$$

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[1] In the formalized Lean kernel, the lower eigenvalue bound $\lambda_{\text{min}}(\mathcal{P}) \geq 1$ from the Stein base term $\mathcal{P} \succeq I$ is used directly, so $\|z\|^2 \leq \mathcal{V}(z)$ and the comparison factor $\sqrt{\lambda_M/\lambda_{\text{min}}(\mathcal{P})}$ reduces to $\sqrt{\lambda_M}$. Rayleigh–Ritz for $\lambda_{\text{min}}(\mathcal{P})$ (Horn and Johnson 2013) is not required.

The parenthesis tends to 1 as $\|z\|_2 \to 0$. Isolating the scalar inequality $1 - at - bt^2 \geq 1/2$ on a compact interval $[0, \delta_\star]$ (with $a = 2\kappa\lambda_M C_R$ and $b = \lambda_M C_R^2$) yields $\delta_\star > 0$ such that $\|z\|_2 \leq \delta_\star$ implies the parenthesis is at least $1/2$. Set

$$\delta_\Omega := \frac{\rho}{\sqrt{1 + (\Delta T)^2}}, \qquad r_0 := \min\{\delta_\Omega, \delta_\star\}.$$

If $\mathcal{V}(z) \leq r_0^2$, then $\|z\|_2 \leq \sqrt{\mathcal{V}(z)} \leq r_0 \leq \min\{\delta_\Omega, \delta_\star\}$. In particular $\|\delta w\|_2 \leq \|z\|_2 < w^\star_{\min}$, so $w^\star + \delta w \in \Omega$ and $\mathcal{E} \subset \Omega \times \mathbb{R}^n$; and $\|w_{t+1} - w^\star\|_2 \leq \rho < w^\star_{\min}$, so $w_{t+1} \in \overline{B}_\rho(w^\star) \subset \Omega$ and the remainder expansion applies. Therefore

$$\mathcal{V}(z_{t+1}) \leq q\,\mathcal{V}(z_t), \qquad q = 1 - \tfrac{1}{2}\lambda_M^{-1}.$$

(Here $\lambda_M \geq 1$, so $q \in (0,1)$.) Forward invariance of $\mathcal{E}$ follows by induction on $t \in \mathbb{N}$, and iterating yields $\mathcal{V}(z_t) \leq q^t \mathcal{V}(z_0)$. The Euclidean bound is the comparison $\|z\|_2^2 \leq \mathcal{V}(z) \leq \lambda_M \|z\|_2^2$, hence

$$\|z_t\|_2 \leq \sqrt{\lambda_M}\, q^{t/2} \|z_0\|_2.$$

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11. Lean 4

The formalization is `lean/`. It does not import RRT. The development is in Lean 4 (Moura and Ullrich 2021) against Mathlib (The mathlib Community 2020).

```
./scripts/verify.sh
```

This fetches the Mathlib cache, builds only `UniversalStability`, and runs `lean/check_axioms.lean`, which **fails** if a kernel LAW depends on `sorryAx` or on any axiom outside {propext, Classical.choice, Quot.sound}. The same script is the GitHub Actions job.

The default library contains no `sorry`. The pipeline Definitions $\to$ Theorem 5 is machine-checked. Discharged statements:

Paper	Lean
Theorem 1 ($DF_c = \mathcal{H}$)	`onShellForceOnShell_hasFDerivAt_of_connected`
Theorem 2 (Hessian floor)	`hessian_quad_ge_floor`, `universal_equilibrium_hessian_posdef`
Theorem 2A (global minimizer)	`exists_minimizer_on_orthant_force_zero`
Theorem 2B (uniqueness)	`theorem2B_unique_on_shell_equilibrium`
Theorem 3 (Jury / modal Schur)	`theorem3_jury`, `theorem3_block_schur`
Theorem 4 ($\rho(\mathcal{J}) < 1$, Stein $\mathcal{P}$)	`theorem4_spectralRadius_lt_one`, `theorem4_stein_lyapunov`
Theorem 5 (ellipsoid invariance)	**`theorem5_local_invariance`**
Non-vacuity (2-node path)	`twoNode_theorem5`

Supporting lemmas for Theorem 5 include `quadratic_margin_half` (scalar polynomial $1 - at - bt^2 \geq 1/2$), `remainder_euc_le_CR` (Taylor remainder composed with the kinematic step, $\sqrt{|E|}$ converting Pi sup-norm to Euclidean length), and `lyap_step_contract` (one-step Stein contraction). Euclidean comparison uses the Stein floor $\mathcal{P} \succeq I$ so $\mathrm{euc}(z)^2 \leq \mathcal{V}(z) \leq \lambda_M\, \mathrm{euc}(z)^2$ with $\lambda_M = \|P\|_2$.

A two-vertex, one-edge path (`UniversalStability.Examples.TwoNodeExample`) inhabits the hypotheses of Theorem 5 at $c = 0$ (force balance reduces to $V'(w^\star) = 0$). The unique positive root is admissible; $w^\star = 1$ is *not* a root, since $V'(1) = -6$. Instantiating `theorem5_local_invariance` produces an $r_0 > 0$, so the hypotheses do not imply `False`.

Incomplete leftovers (finite-dimensional mountain pass (Ambrosetti and Rabinowitz 1973); Hessian $\Rightarrow$ `IsLocalMin`) live in `USIncomplete` and are not cited as discharging any statement above.

See `lean/README.md` for pins and module layout.

The manuscript, Lean sources, CI, and all other files in this repository were generated by AI under human direction.

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